



PROJECT REPORT

ON

CHESSMASTER: A Web-Based Multiplayer and Analysis Platform

Bachelor of Technology
in
Computer Science and Engineering

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Project Overview:

ChessMaster is a full-stack web-based chess application developed for both casual and competitive players. It allows users to play chess in two modes: against AI and in real-time multiplayer. The system also includes a powerful game analysis tool that utilizes the Stockfish engine to evaluate moves and provide feedback. Users can view and track their game statistics, including wins, losses, and ratings. A logging mechanism records important user activities such as login, logout, and game outcomes. The platform features a clean, responsive UI designed for professional usability. It ensures accessibility by allowing analysis without requiring login. Security measures such as session handling and password hashing are implemented. Tutorials are also included to help beginners improve their gameplay. The project showcases the integration of frontend, backend, database, and AI components into one cohesive system.

INITIAL PROPOASL:

Chess Platform Software Description

a. Purpose

The purpose of this software is to provide an interactive platform for playing chess. It caters to users ranging from beginners to advanced players by offering real-time multiplayer games, AI-powered matches, tutorials, and performance analytics. The platform aims to enhance chess learning and playing experiences through an intuitive interface and robust features.

b. Users

1. **Casual Players:** Users who want to play chess for recreation.
2. **Chess Enthusiasts:** Individuals seeking to improve their chess skills through AI matches or tutorials.
3. **Competitive Players:** Players interested in real-time multiplayer games and tournaments.
4. **Educators/Students:** Chess trainers and learners who want a structured platform for learning chess strategies.

c. Technology to be Used

- **Frontend:**
 - HTML, CSS, JavaScript (with React.js for dynamic interfaces)
 - Chessboard.js for chessboard visualization
- **Backend:**
 - Node.js with Express.js for managing APIs and server-side logic
 - WebSocket (Socket.IO) for real-time gameplay
- **Database:**
 - MySQL (for relational data management)
- **AI/Chess Engine:**
 - Stockfish or chess.js for game logic and AI moves
- **Hosting:**
 - Cloud platform like AWS or free hosting platforms (GitHub Pages for frontend and Heroku for backend)

d. Estimated Lines of Code

- Frontend: ~1000 lines
- Backend: ~1000 lines
- AI Integration: ~500 lines
- Testing and Deployment Scripts: ~500 lines

- **Total Estimated Lines of Code:** ~3000 lines

e. Five Functionalities

1. Real-Time Multiplayer Gameplay

- **Input:** Player 1 moves a piece (e.g., e2 to e4)
- **Output:** Move is reflected on both players' boards in real time. Game state updates (e.g., valid move or check notification)

2. Play Against AI

- **Input:** User selects AI difficulty level and makes a move
- **Output:** AI responds with the best possible move based on difficulty. Game continues until checkmate or draw

3. Game Analysis

- **Input:** User uploads a past game file in Portable Game Notation (PGN) format
- **Output:** Software analyzes the game, highlights mistakes, and suggests alternative moves

4. Chess Tutorials

- **Input:** User selects a tutorial (e.g., "Basic Opening Principles")
- **Output:** Interactive guide with text, visuals, and exercises to teach the selected topic

5. Leaderboard and Player Statistics

- **Input:** User completes games, earning points based on performance
- **Output:** Updates user statistics (e.g., win/loss ratio, Elo rating) and displays global/local leaderboards

f. Software Delivery Method

This software will be delivered as a **web service**:

- Accessible via browsers for real-time multiplayer and learning features.

Software Requirements Specification (SRS)

1. Introduction

Goal

The goal for this software is to provide a platform to the users to play multiplayer games, games against ai, analysis of game, daily puzzles and also the performance evaluation. This platform will help the players to improve their chess skills using ai powered matches and game analysis also help beginners to learn the basics of the game using tutorials.

Stakeholders and Users

Stakeholders:

- Developers
- System Administrators
- Chess Trainers/Institutions

Users:

- **Casual Players** – Play for recreation.
 - **Chess Enthusiasts** – Improve skills through AI matches and tutorials.
 - **Competitive Players** – Participate in real-time multiplayer games and tournaments.
 - **Educators/Students** – Learn and teach chess using structured tutorials.
-

2. Functional Requirements

Features and Expected Behaviours

1. Real-Time Multiplayer Gameplay

- **Input:** Player 1 moves a piece (e.g., e2 to e4).
- **Output:** The move is reflected on both players' boards in real-time.
- **Behaviour:** The system validates the move, updates the game state, and notifies the opponent.

2. Play Against AI

- **Input:** User selects AI difficulty and makes a move.
- **Output:** AI responds with an optimal move.

- **Behaviour:** The AI engine (Stockfish) calculates and returns the best move.

3. Game Analysis

- **Input:** User uploads a Portable Game Notation (PGN) file.
- **Output:** The system highlights mistakes and suggests improvements.
- **Behaviour:** AI evaluates the game and provides insights.

4. Chess Tutorials

- **Input:** User selects a tutorial topic.
- **Output:** Interactive guide with text, visuals, and exercises.
- **Behaviour:** The system tracks progress and provides feedback.

5. Leaderboard and Player Statistics

- **Input:** Players complete matches.
- **Output:** Updated leaderboards and statistics (e.g., win/loss ratio, Elo rating).
- **Behaviour:** The system records game results and ranks users accordingly.

3. Non-Functional Requirements

Performance

- **Response Time:** Moves should be reflected in real-time (<500ms latency for multiplayer games).
- **Throughput:** The system should support up to 1000 concurrent users.

Security Considerations

- Secure authentication using JWT for user logins.
- Encrypted communication via HTTPS.
- Prevention of cheating (anti-bot detection, fair play verification).

Usability & Accessibility

- Responsive UI.
- Keyboard and touch-friendly navigation.

Reliability & Availability

- 99.9% uptime guarantee.
- Auto-reconnect feature in case of network disconnection.

Format

- Chessboard displayed using Chessboard.js.
- Backend in Node.js (Express.js) with MySQL database.

4. Constraints and Limitations

Time and Budget Constraints

- The project should be developed within **five weeks**.
 - Limited budget for hosting (use free/affordable cloud platforms like Heroku for backend and GitHub Pages for frontend).
-

5. Acceptance Criteria

- Functional features must work as expected without critical bugs.
 - AI should provide reasonable moves for different difficulty levels.
 - Multiplayer functionality should allow smooth real-time interaction.
 - Tutorials must be interactive and engaging.
 - Leaderboards should correctly rank users based on their performance.
-

6. Time Frame / Roadmap

Five-Week Development Plan

- **Week 1:** Setup project, database schema, and authentication.
 - **Week 2:** Implement chessboard UI and AI engine.
 - **Week 3:** Develop real-time multiplayer gameplay.
 - **Week 4:** Integrate game analysis, tutorials, and leaderboards.
 - **Week 5:** Testing, bug fixes, and deployment.
-

7. References

- [Stockfish Chess Engine](#)
 - [Chess.js Documentation](#)
 - [Chessboard.js Documentation](#)
-

8. API Used

- **Multiplayer Gameplay API – WebSocket with Socket.IO**
 - **AI Gameplay API – Stockfish/Chess.js**
-

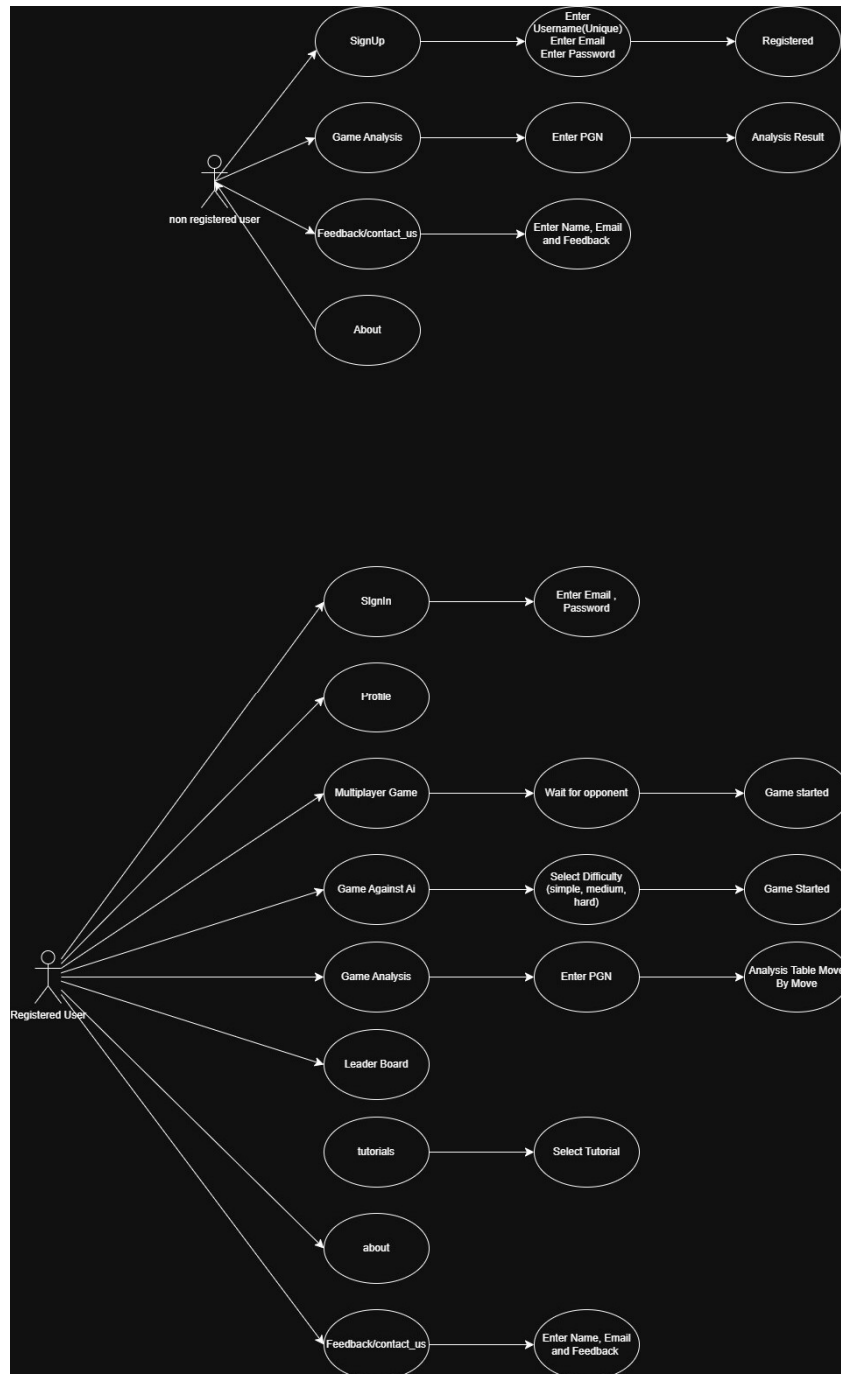
9. Additional Information

Written By: Abhishek Jangir

Version: 1.0

Date: 5-February-2025

Use Case Diagram:



◆ User Stories – *ChessMaster* Web Application

1. User Registration

As a new visitor,

I want to create an account on the platform

So that I can securely access personalized features and track my progress.

2. User Authentication

As a returning user,

I want to log in using my credentials

So that I can resume my activities and maintain my profile and ratings.

3. AI Mode Gameplay

As a casual player,

I want to play against an AI opponent

So that I can practice and improve my gameplay at my own pace.

4. Multiplayer Chess Matches

As a competitive user,

I want to play real-time chess games with other online users

So that I can challenge myself and improve my ranking.

5. Leaderboard Access

As a registered user,

I want to view a global leaderboard

So that I can compare my performance with other players.

6. Game Analysis

As a chess learner,

I want to upload PGN files and analyze completed games

So that I can evaluate move quality and understand strategic improvements.

7. Activity Logs

As a logged-in user,

I want to view my activity logs

So that I can review actions such as logins, gameplay, and interactions.

8. Public Tutorial Access

As a visitor or learner,

I want to access chess tutorials without needing to log in

So that I can explore and learn about the game before registering.

9. Contact & Support

As a user,

I want to submit queries or feedback through a contact form

So that I can communicate with the support team for assistance.

Use Case 1: User Registration

- **Actor:** Guest User
 - **Preconditions:** User is on the registration page
 - **Main Flow:**
 1. User enters name, email, password, and confirms password
 2. System validates the inputs
 3. System sends an OTP to the user's email for verification
 4. User enters the OTP
 5. System verifies OTP and creates a new account
 6. User is redirected to the login page
 - **Postcondition:** Account is successfully created
-

Use Case 2: User Login

- **Actor:** Registered User
 - **Preconditions:** User has a valid account
 - **Main Flow:**
 1. User enters email and password
 2. System authenticates credentials
 3. If successful, user session is created
 4. User is redirected to the dashboard
 - **Alternate Flow:**
 - If credentials are incorrect, system shows error message
 - **Postcondition:** User is logged in and redirected to dashboard
-

Use Case 3: Play Against AI

- **Actor:** Logged-in User

- **Preconditions:** User is authenticated
 - **Main Flow:**
 1. User clicks on “Play Against AI”
 2. System loads the chessboard interface with the AI logic
 3. User plays as white; AI plays as black
 4. Game continues until checkmate, draw, or resignation
 5. Game result and updated stats are shown
 - **Postcondition:** Game ends, and stats are updated
-

Use Case 4: Play Multiplayer Game

- **Actor:** Logged-in User
 - **Preconditions:** Two users are online and available to play
 - **Main Flow:**
 1. User enters multiplayer mode
 2. System matches the user with another available player
 3. Game begins with both players
 4. System handles turn-based moves
 5. Game ends with a result
 6. System updates win/loss/draw stats for both players
 - **Postcondition:** Match result recorded, stats updated
-

Use Case 5: Analyze a Game

- **Actor:** Guest or Logged-in User
- **Preconditions:** User has a valid PGN file
- **Main Flow:**
 1. User visits the analysis page

2. User uploads a PGN file
 3. System parses the PGN and shows the move-by-move analysis
 4. Evaluation bar and best moves are displayed
- **Postcondition:** User sees analysis of the uploaded game
-

Use Case 6: View Leaderboard

- **Actor:** Any User
 - **Preconditions:** None
 - **Main Flow:**
 1. User clicks on the “Leaderboard” link
 2. System fetches top 50 users based on rating
 3. Leaderboard is displayed
 - **Postcondition:** User views current top players
-

Use Case 7: Contact Support

- **Actor:** Any User
 - **Preconditions:** User is on the Contact Us page
 - **Main Flow:**
 1. User fills in name, email, and message
 2. System validates input
 3. System stores the message in the database
 4. Confirmation is shown to the user
 - **Postcondition:** User message is recorded for admin review
-

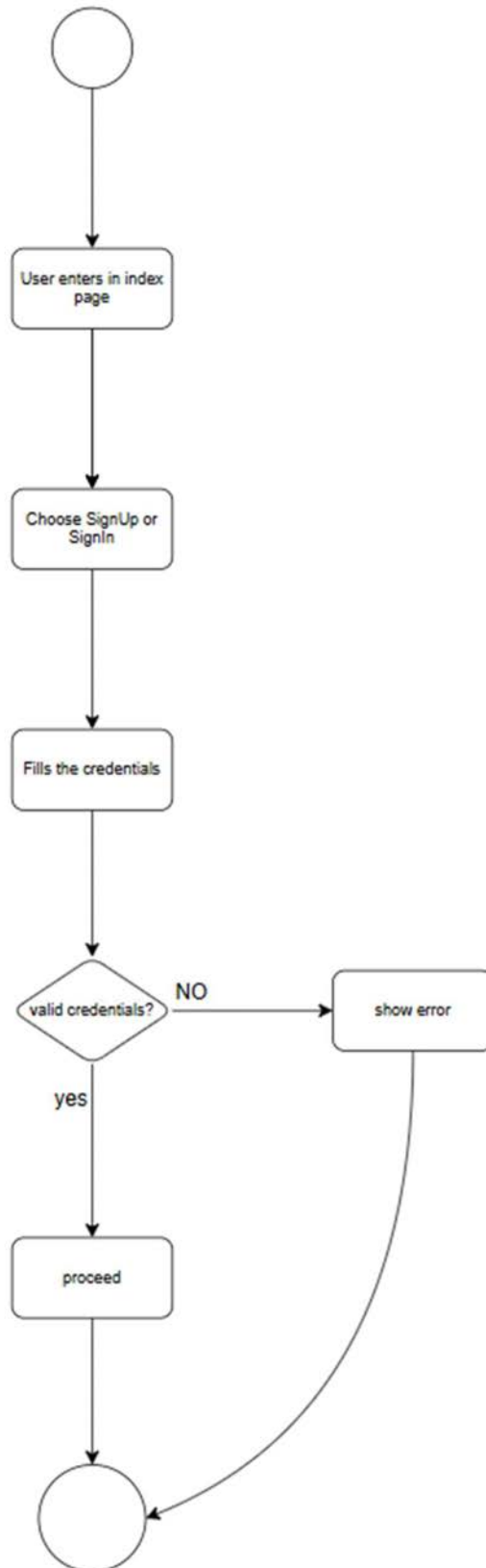
Use Case 8: View Logs

- **Actor:** Logged-in User

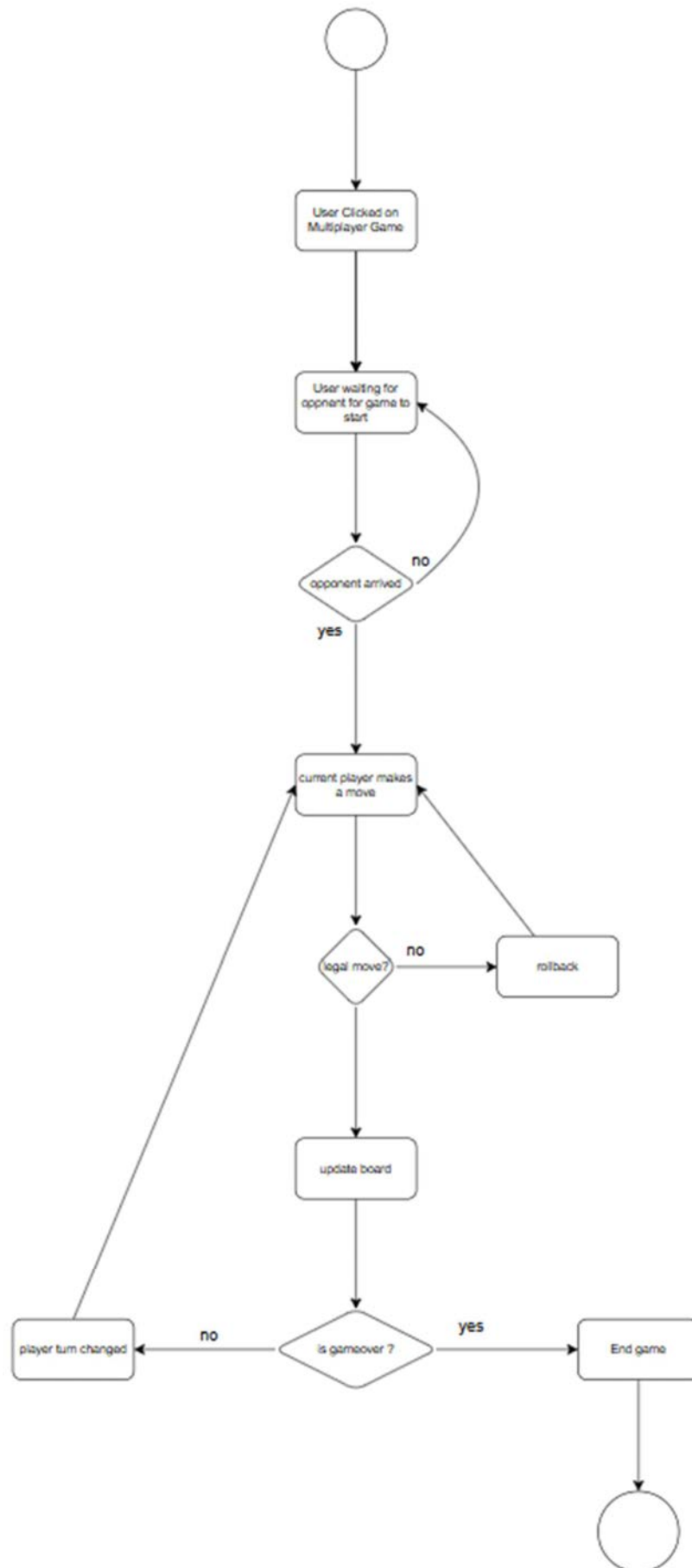
- **Preconditions:** User is authenticated
- **Main Flow:**
 1. User navigates to “Logs” page
 2. System fetches and displays their activity logs
- **Postcondition:** User sees their session history

Activity and Data Flow Diagrams

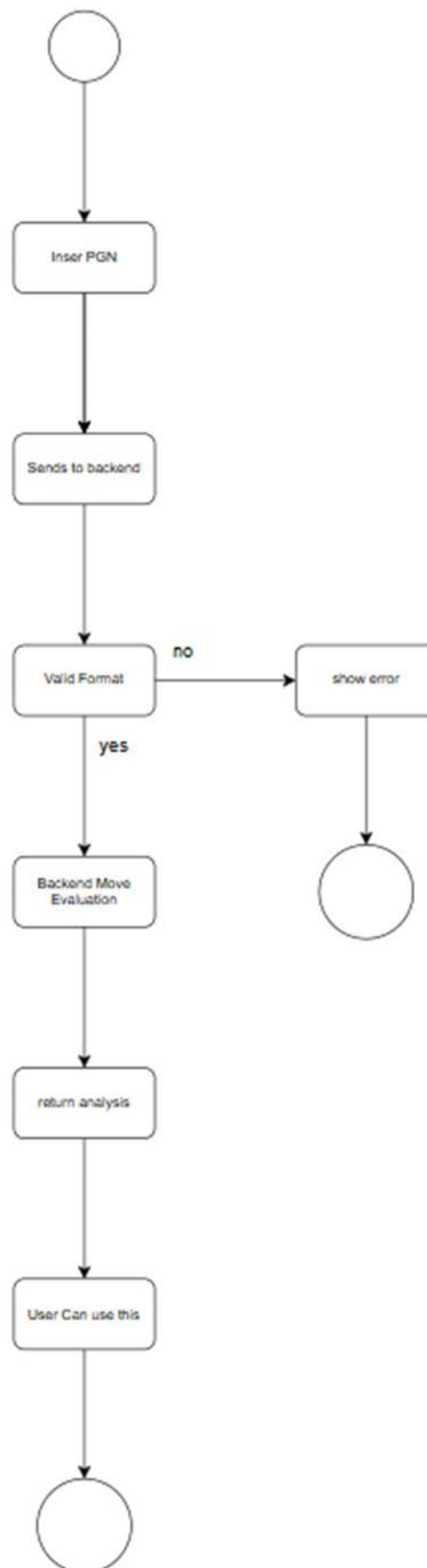
Activity Diagram for Login Registration



Multiplayer Chess:

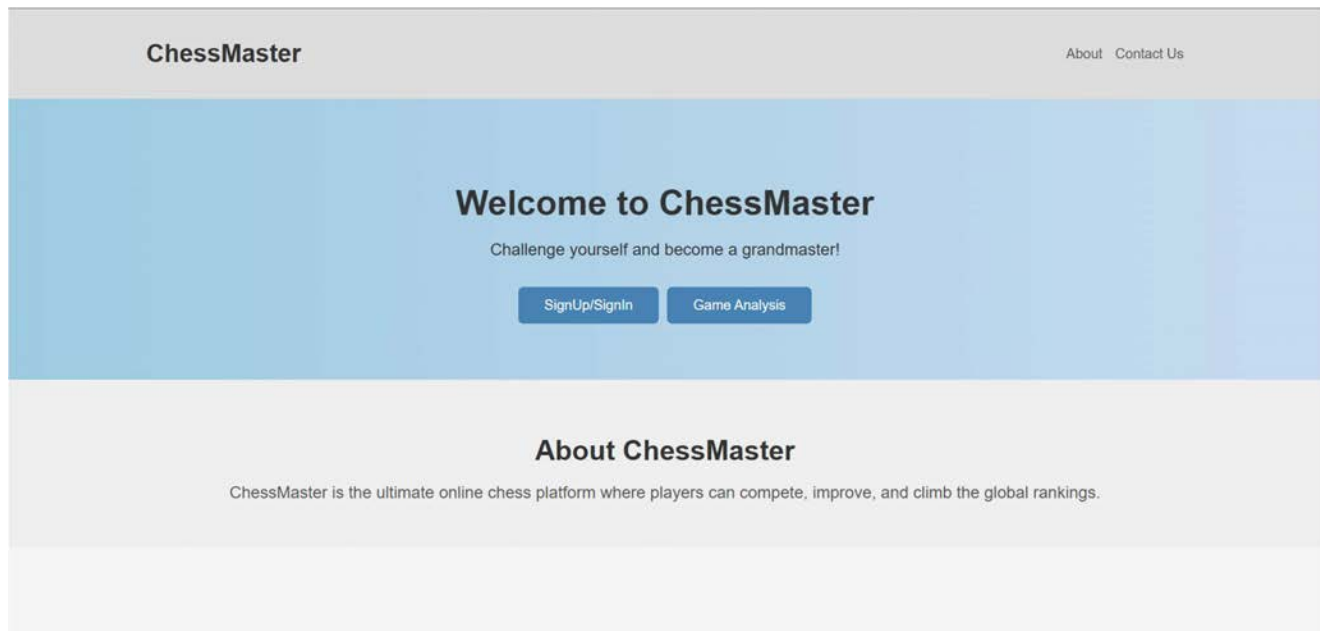


PGN Analysis:

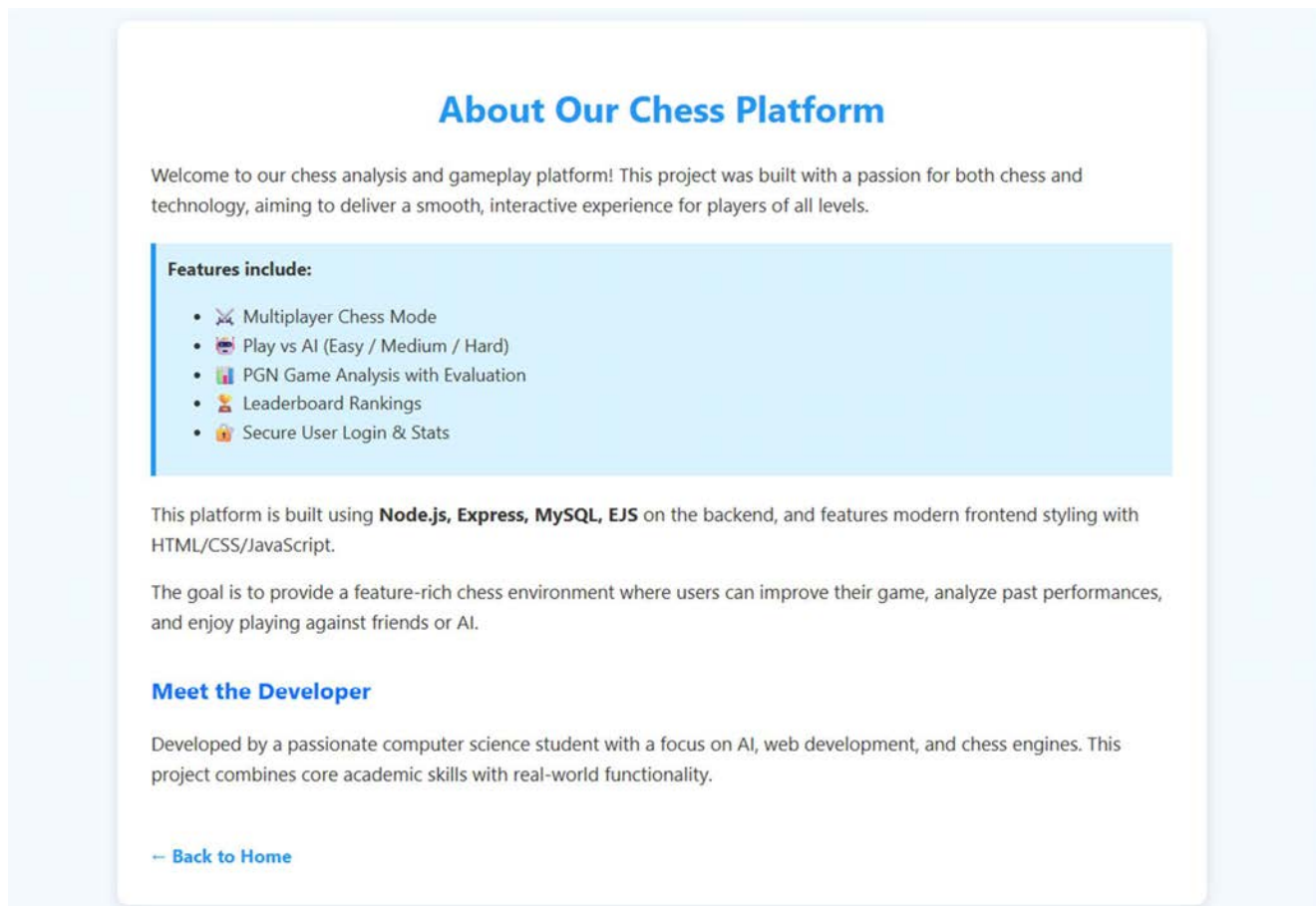


Release 1 Notes:

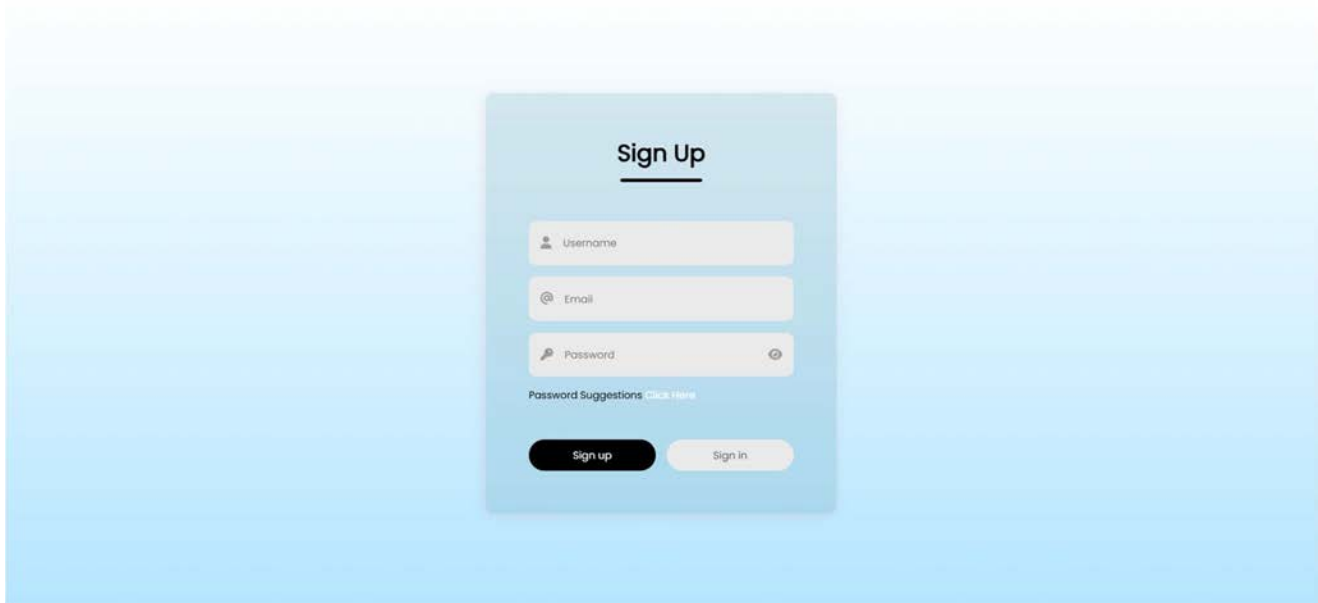
Index page:



About page:



Sign Up Page:



The Sign Up form is centered on a blue gradient background. It features a light blue rounded rectangle with a subtle shadow. The title "Sign Up" is at the top, underlined. Below it are three input fields: "Username" with a person icon, "Email" with an @ icon, and "Password" with a key icon and a toggle eye icon. A link "Click Here" is next to the password field. At the bottom are two buttons: "Sign up" (black with white text) and "Sign in" (white with black text).

Sign Up

Username

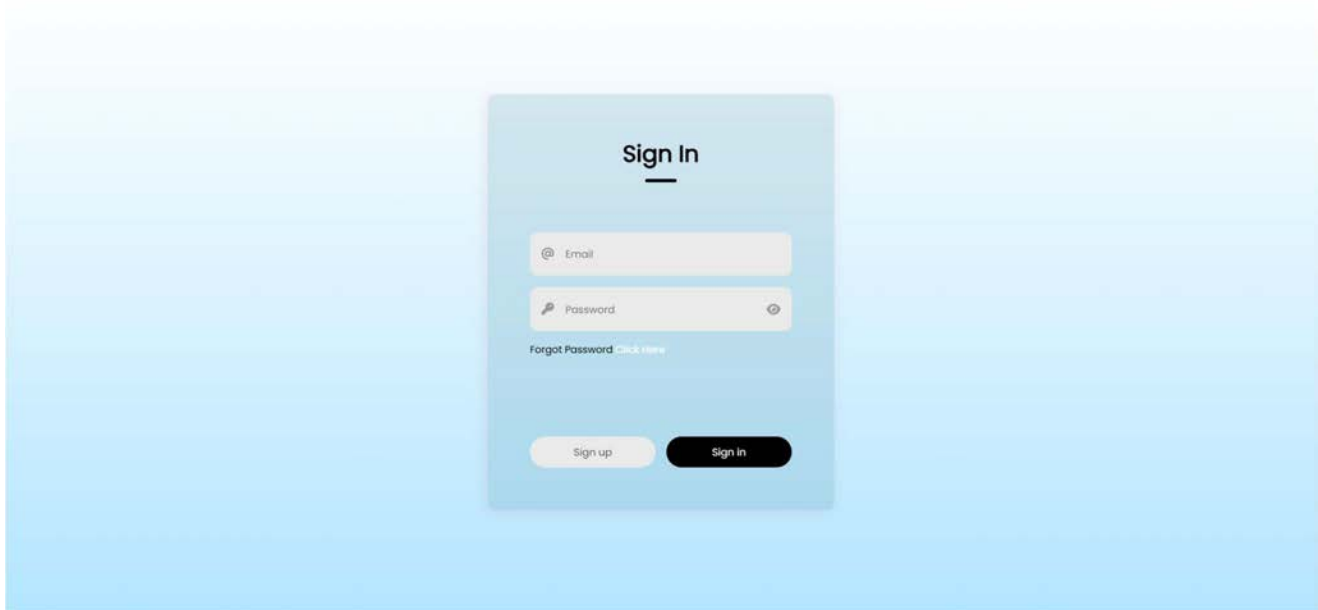
Email

Password

[Click Here](#)

Sign up Sign in

Sign In Page:



The Sign In form is centered on a blue gradient background. It features a light blue rounded rectangle with a subtle shadow. The title "Sign In" is at the top, underlined. Below it are two input fields: "Email" with an @ icon and "Password" with a key icon and a toggle eye icon. A link "Click Here" is next to the password field. At the bottom are two buttons: "Sign up" (white with black text) and "Sign in" (black with white text).

Sign In

Email

Password

[Click Here](#)

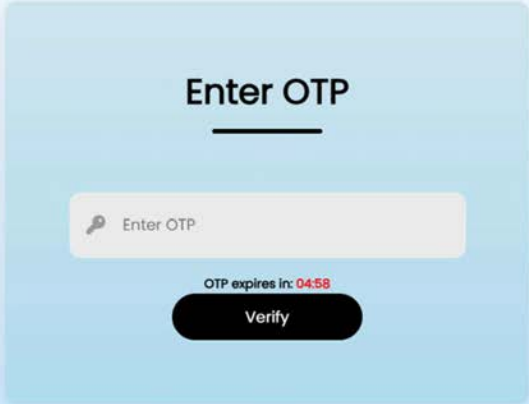
Sign up **Sign in**

Forgot Password:



The image shows a 'Forgot Password' form on a light blue background. The form is a rounded rectangle with a light blue header containing the title 'Forgot Password' and a horizontal line. Below the header is a white input field with a light gray border, containing an '@' icon and the text 'Enter your email'. Underneath the input field is a black rounded button with the word 'Verify' in white text.

Enter Otp:



The image shows an 'Enter OTP' form on a light blue background. The form is a rounded rectangle with a light blue header containing the title 'Enter OTP' and a horizontal line. Below the header is a white input field with a light gray border, containing a key icon and the text 'Enter OTP'. Underneath the input field is a red text label 'OTP expires in: 04:58'. At the bottom is a black rounded button with the word 'Verify' in white text.

Create New Password:

Create New Password

 New Password 

 Confirm Password 

Password Suggestions [Click Here](#)

Submit

Home page:

ChessMaster

ProfileLeaderboardAboutContact UsLogsLogout

Welcome to ChessMaster

Challenge yourself and become a grandmaster!

Play ▾

Game Analysis

Tutorials

Your Chess Journey Starts Here

Welcome to **ChessMaster** — where strategy meets simplicity. Challenge yourself, play with AI or real opponents, and improve every game.

Logged in as: 17

Profile Page:





View

Change

Username: AbhishekJangiraj

Email: abhishekjangir1234aj@gmail.com

Total Matches Played

3

Matches Won

1

Matches Lost

1

Matches Drawn

1

Rating

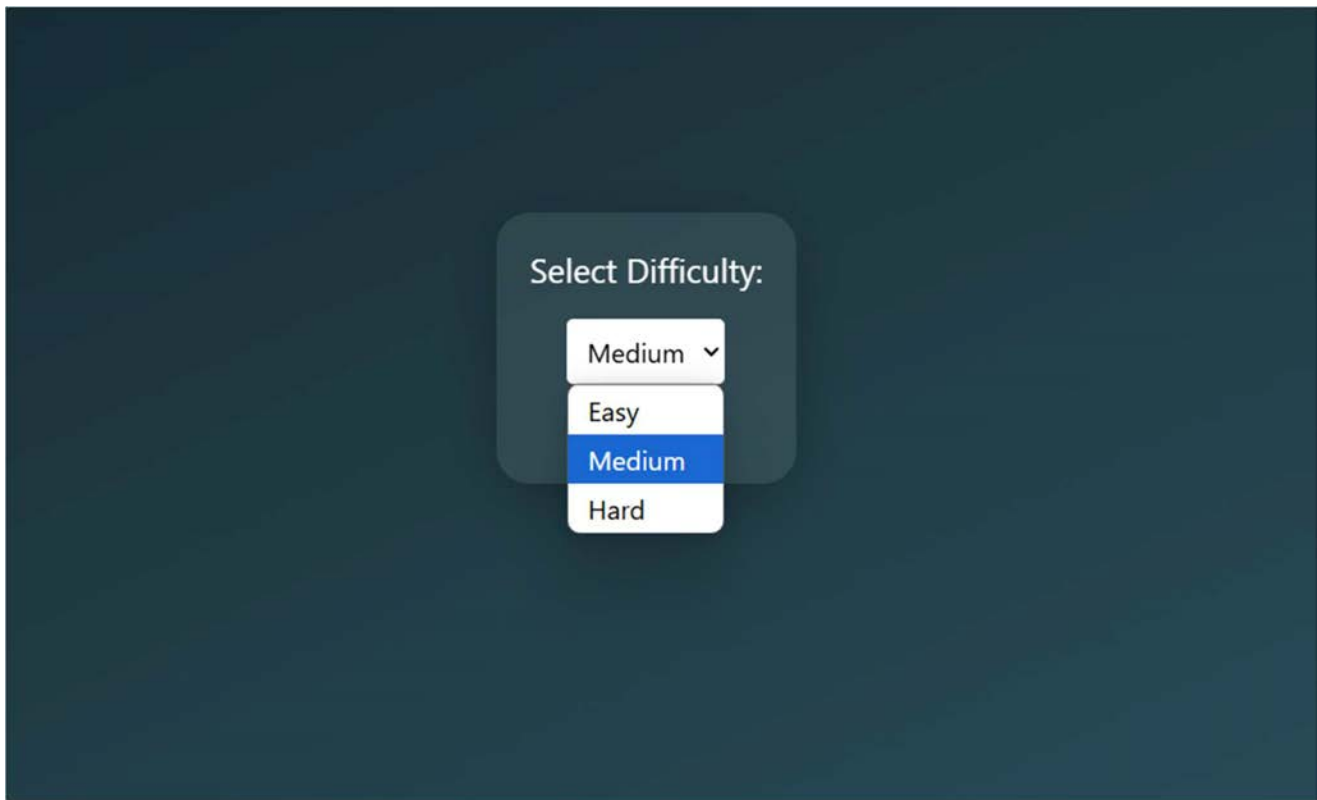
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Multiplayer Game:



Game Against AI:

→ Difficulty selection



Q1. What have you done?

I have completed the following core components of the ChessMaster platform:

- Fully functioning signup, login, forgot-password and logout system using sessions.
- Play Against AI mode integrated with Stockfish engine.

- Multiplayer mode allowing real-time gameplay between two users.
- A working profile page showing the logged-in user's data and stats.
- A themed and styled about page.
- All views rendered using EJS with consistent layout and responsive design.

Q2. What is the target for the next increment (within 10 days)?

- Add Game Analysis feature: upload PGN and analyze moves using Stockfish.
- Create a Chess Tutorials page for learners.
- Implement a proper user logs viewer interface (already logging events).
- Implementation of feedback form.

Q3. Any difficulties?

Yes, I faced a few real challenges:

- Chess.js Version Conflict: Downgrading chess.js caused issues in the AI gameplay. We had to debug PGN loading and ensure compatibility with our move logic.
- WebSocket Communication Issues

Setting up and synchronizing WebSocket communication between two players in multiplayer mode was one of the toughest parts. We struggled with:

- Player disconnection and reconnection handling.
- Ensuring turn-based logic stayed consistent.
- Broadcasting moves only to the correct opponent. These were eventually handled using socket.io namespaces and proper user mapping.
- Stat Updates in Multiplayer: Ensuring consistent updating of win/loss/draw records and ratings post-game required session handling and accurate game-end detection.
- No External CSS Policy: Since all styles had to be inside EJS without external CSS, maintaining consistent and responsive UI styling across all pages became more tedious.
- Session-Based Routing: Preventing unauthorized access to protected routes while maintaining game state was tricky and required fine-tuning route middlewares.

Github repolink : <https://github.com/Abhishek7Jangir/Chess2.com>

README for Release 1 – ChessMaster

Project Name: ChessMaster – Web-Based Chess Platform

Working Features in Release 1

1. User Authentication

- Signup and login functionality using secure sessions.
- Logout functionality clears user sessions properly.

2. Play Against AI

- A functional chessboard that lets users play against a Stockfish-powered AI.
- The user always plays white; the difficulty is set to medium.
- No drag-and-drop required—moves are click-based.

3. Multiplayer Chess

- Two-player real-time gameplay.
- Automatic turn-based system.
- Game results update win/loss/draw and ratings.
- Session-based matchmaking system.

4. Profile Page

- Displays username, profile photo, and match statistics.
- Handles default image fallback.

5. Leaderboard

- Shows top 50 users by rating.
- Displays total matches, wins, losses, and draws.

6. About Section

- Overview of the platform's mission and goals.

7. Logout

- Clears user session securely.
- Logs the logout action.

8. Responsive Dashboard

- Sidebar with navigation (collapsible on mobile).
- Buttons to play AI/multiplayer or access other pages.
- Matching color theme consistent across pages.

ChessMaster - Release Test Cases

Signup Module

TEST CASE ID	TEST SCE-NARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	POST CONDI-TION	ACTUAL RESULT	STATUS (PASS/FAIL)
TC_SIGNUP_001	Verify signup with valid input	Enter valid name, email, password	User not signed up	1. Enter name 2. Enter email 3. Enter password 4. Click "Signup"	Valid name, email, password	Account created, redirected to login	User account created	As expected	Pass
TC_SIGNUP_002	Missing fields on signup	Leave one or more fields blank	-	1. Leave field blank 2. Click "Signup"	Empty fields	Error: "All fields required"	Remains on signup page	As expected	Pass
TC_SIGNUP_003	Duplicate email signup	Use already registered email	Email exists	1. Enter duplicate email 2. Click "Signup"	Existing email	Error: "Email already in use"	Remains on signup page	As expected	Pass
TC_SIGNUP_004	Weak password check	Enter weak password	-	1. Input weak password 2. Click "Signup"	Weak password	Error: "Password not strong"	Remains on signup page	As expected	Pass

Login Module

TEST CASE ID	TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	POST CONDITION	ACTUAL RESULT	STATUS (PASS/FAIL)
TC_LOGIN_001	Valid login check	Enter valid credentials	User registered	1. Enter email 2. Enter password 3. Click "Login"	Valid email/password	Redirect to dashboard	Dashboard shown	As expected	Pass
TC_LOGIN_002	Wrong password check	Use correct email, wrong password	User registered	Same steps as above	Invalid password	Error: "Invalid credentials"	Remains on login page	As expected	Pass
TC_LOGIN_003	Unregistered email	Use unregistered email	-	Same steps as above	Unregistered email	Error: "Invalid credentials"	Remains on login page	As expected	Pass
TC_LOGIN_004	Empty fields	Leave email/password blank	-	1. Leave blank 2. Click "Login"	Blank fields	Error: "Please enter all fields"	Remains on login page	As expected	Pass

Logout Module

TEST CASE ID	TEST SCE-NARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	POST CONDI-TION	ACTUAL RESULT	STATUS (PASS/FAIL)
TC.LOGOUT_001	Valid logout	Click logout button	User logged in	1. Click "Logout"	-	Session destroyed, redirect to login	User logged out	As expected	Pass
TC.LOGOUT_002	Logout without login	Access logout URL directly	User not logged in	1. Visit logout route	-	Redirected to login	Login page shown	As expected	Pass

Play Against AI Module

TEST CASE ID	TEST SCE-NARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	POST CONDI-TION	ACTUAL RESULT	STATUS (PASS/FAIL)
TC_AI.001	Load AI board	Visit AI play page	User logged in	1. Click "Play AI"	-	AI board with white pieces loads	AI board shown	As expected	Pass
TC_AI.002	Make valid move	Move e2 to e4	Game started	1. Move e2-e4	Legal move	AI responds	Game continues	As expected	Pass
TC_AI.003	Try invalid move	Move e2 to e5	Game started	1. Move e2-e5	Invalid move	Move blocked or ignored	Same board state	As expected	Pass
TC_AI.004	Refresh page during game	Refresh mid-game	Game ongoing	1. Play 2. Refresh page	-	Game resets or resumes	New/continued board	As expected	Pass

Multiplayer Game Module

TEST CASE ID	TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	POST CONDITION	ACTUAL RESULT	STATUS (PASS/FAIL)
TC_MP_001	Start multiplayer	Initiate game with other user	2 players online	1. Click "Multiplayer"	-	Game starts with opponent	Game started	As expected	Pass
TC_MP_002	Opponent disconnects	Opponent closes browser	Game in progress	1. Close opponent tab	-	Message "opponent disconnected"	Game ends or paused	As expected	Pass
TC_MP_003	Move sync	Player makes move	Game in progress	1. Player moves 2. Check opponent board	e2-e4	Move appears live for both players	Turn switches	As expected	Pass
TC_MP_004	Game end condition	Checkmate move	Game in progress	1. Deliver checkmate	Valid checkmate	Game ends, result shown	Game over screen	As expected	Pass

Profile Module

TEST CASE ID	TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	POST CONDITION	ACTUAL RESULT	STATUS (PASS/FAIL)
TC_PR_001	View profile	Access profile page	User logged in	1. Visit /profile	-	User info and photo shown	Profile loaded	As expected	Pass
TC_PR_002	Upload valid photo	Upload JPG/PNG ; 5MB	-	1. Go to profile 2. Upload image	valid.jpg (3MB)	Image updated	Image displayed	As expected	Pass
TC_PR_003	Upload unsupported file	Upload .exe/.pdf	-	1. Upload file	test.exe	Error message shown	Upload blocked	As expected	Pass
TC_PR_004	Upload oversized image	>5MB image	-	1. Upload large image	big.jpg (7MB)	"File too large" message	Upload blocked	As expected	Pass
TC_PR_005	Access without login	Go to /profile	User not logged in	1. Visit /profile	-	Redirect to login	Login page shown	As expected	Pass
TC_PR_006	Default profile image	No photo uploaded	-	1. Go to profile	-	Placeholder shown	Placeholder shown	As expected	Pass

About Page Module

TEST CASE ID	TEST SCE-NARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	POST CONDI-TION	ACTUAL RESULT	STATUS (PASS/FAIL)
TC_AB_001	View about page	Access via menu	Any user	1. Click "About"	-	Static info page loads	About shown	As expected	Pass
TC_AB_002	Access without login	Visit /about directly	Not logged in	1. Visit /about	-	About page loads	Static content shown	As expected	Pass

Leaderboard Module

TEST CASE ID	TEST SCE-NARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	POST CONDI-TION	ACTUAL RESULT	STATUS (PASS/FAIL)
LB_01	View leaderboard as logged-in user	Open leaderboard page	User logged in	1. Log in 2. Click "Leaderboard"	-	Leaderboard with player names and ranks loads	Leaderboard visible	As expected	Pass
LB_02	View leaderboard as guest	Access leaderboard without login	Not logged in	1. Open "/leaderboard" route	-	Leaderboard is not visible or redirected	Remains on same or redirected page	As expected	Pass
LB_03	Verify ranking order	Ensure correct sorting	User logged in	1. Open leaderboard 2. Observe order	-	Players sorted in descending order of wins/points	Rankings visible correctly	As expected	Pass

Contact Us Module

TEST CASE ID	TEST SCE-NARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	POST CONDI-TION	ACTUAL RESULT	STATUS (PASS/FAIL)
CU_01	Submit valid contact form	Fill contact form completely	-	1. Enter name, email, message 2. Click Submit	Valid form data	Message sent and confirmation shown	Contact message stored/sent	As expected	Pass
CU_02	Submit form with empty fields	Leave fields blank	-	1. Leave one or more fields blank 2. Click Submit	Blank fields	Error: "All fields are required"	Remains on contact form	As expected	Pass
CU_03	Submit with invalid email	Enter invalid email	-	1. Type invalid email format 2. Click Submit	test@com or test	Error: "Enter a valid email address"	Remains on contact form	As expected	Pass

Game Analysis Module

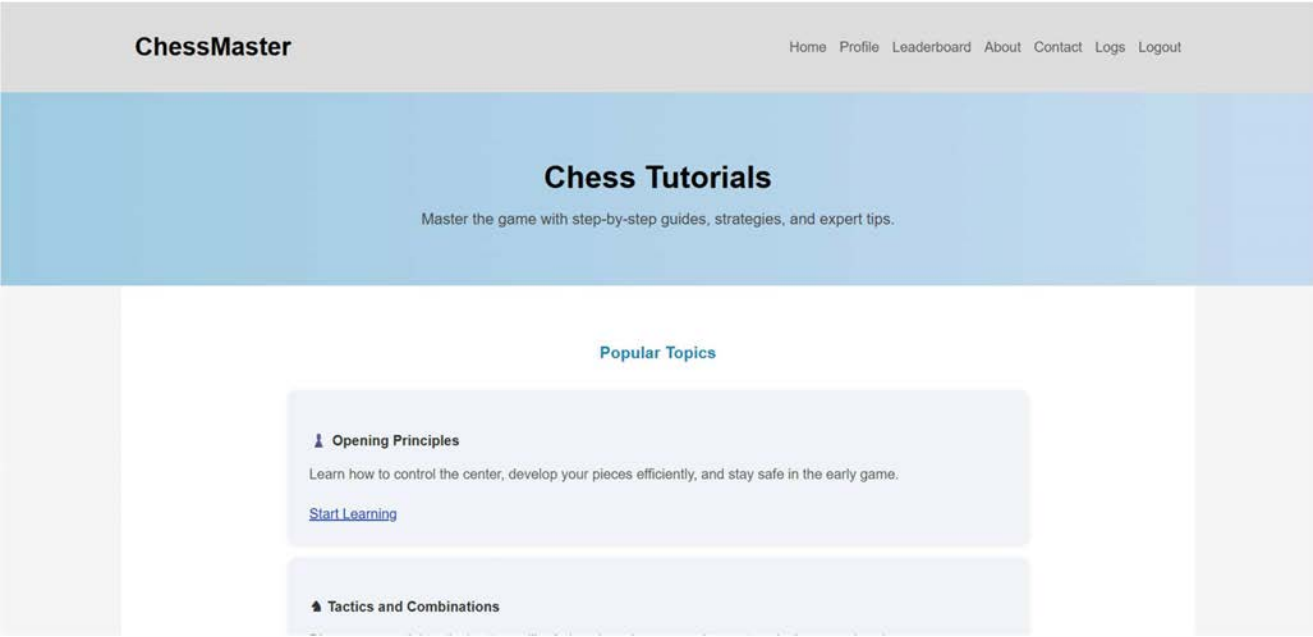
TEST CASE ID	TEST SCENARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	POST CONDITION	ACTUAL RESULT	STATUS (PASS/FAIL)
GA_01	Submit valid PGN	Upload correct PGN	-	1. Choose .pgn file 2. Click Submit	Valid PGN file	Game analysis with best moves displayed	Analysis visible	As expected	Pass
GA_02	Submit invalid PGN	Upload broken/corrupt PGN	-	1. Upload corrupted or malformed PGN	Corrupted PGN file	Error: "Invalid PGN format"	No analysis performed	As expected	Pass
GA_03	Access analysis without login	Visit analysis page	Not logged in	1. Go to "/analysis"	-	Page should load and work	Analysis interface accessible	As expected	Pass
GA_04	Timeout from Stockfish	Upload long PGN	-	1. Upload PGN with 200+ moves	Long PGN file	Long wait or timeout message displayed	User notified or redirected	As expected	Pass

Chess Tutorials Module

TEST CASE ID	TEST SCE-NARIO	TEST CASE	PRE-CONDITION	TEST STEPS	TEST DATA	EXPECTED RESULT	POST CONDI-TION	ACTUAL RESULT	STATUS (PASS/FAIL)
CT_01	Open tutorial page	Click tutorial link	-	1. Click "Tutorials" from navigation menu	-	Tutorial page opens with learning content	Tutorials available	As expected	Pass
CT_02	Check page responsive-ness	Resize browser window	-	1. Resize screen 2. Test on mobile view	-	Page remains usable and readable	Layout adjusts responsively	As expected	Pass
CT_03	Test internal links	Click on in-page section links	-	1. Click a section or chapter link	-	Page scrolls or jumps to that section	Smooth scroll or jump	As expected	Pass

Release 2 Notes:

Chess Tutorials:



Leader Board:

Leaderboard						
#	Username	Rating	Games	Wins	Losses	Draws
1	Abhisheklangiraj	375		1	1	1
2	Abhishek	300		0	0	0
3	hjbgh	300		0	0	0
4	iodjsfoa	300		0	0	0
5	Ahishek Jan	300		0	0	0
6	Abhishek /\	300		0	0	0
7	VishalSolanki	300		0	0	0
8	Pushpa	290		0	0	0
9	MohitSaini	280		0	1	1
10	Anmol	260		0	0	0

Back to Home

Contact Us Page:

Contact Us

Name:

Email:

Message:

Plax 500 characters...

Maximum 500 characters allowed.

Send Message

User Logs:

My Activity Logs

Event Type	Message	Timestamp
Login	User logged in	14/5/2025, 9:46:07 am
Login	User logged in	14/5/2025, 5:46:17 am
Logout	User logged out	14/5/2025, 4:56:34 am
Login	User logged in	14/5/2025, 4:50:28 am
Login	User logged in	14/5/2025, 4:37:53 am
Logout	User logged out	14/5/2025, 4:29:49 am
Login	User logged in	14/5/2025, 4:01:55 am
Login	User logged in	13/5/2025, 10:45:35 pm
Login	User logged in	13/5/2025, 10:34:57 pm
Login	User logged in	13/5/2025, 10:14:59 pm

Game Analysis:

Analyze Chess Game

Enter Chess PGN:

Playing for the fast h4 was the right way to go)) (12... Nf6 13. Rh4) 13. Nf3 Bd7 (13... Kf8 {Was slightly more resilient}) 14. Bf4 [[pgndiagram] Already it is hard to find a move for Black. Ne5 is a terrible threat} f6 (14... Nf6 15. Ne5) 15. Ne4 (Now Nf2 is on the table as well) Ngh6 \$2 {A blunder in a lost position} (15... 0-0-0 16. Qa5) (15... Kf8 16. Nf2 Qg6 17. g4 [[pgndiagram] And white will win material} Re8 18. Bxf5 Bxf5 19. Nh4) 16. Bxh6 Nxh6 17. Rxh6 \$1 {Winning the piece or the queen} gxh6 18. Nxf6+ Kf8 19. Nxc4 [[pgndiagram] An impressive victory which allowed Steinitz to capture the crown.} 1-0

Analyze Game

Analysis Results		
Move Analysis		
Move	FEN	Best Move
e2e4	rnbqkbnr/pppppppp/8/8/8/PPPPPPPP/RNBQKBNR w KQkq - 0 1	e2e4
e7e5	rnbqkbnr/pppppppp/8/8/4P3/8/PPPP1PPP/RNBQKBNR b KQkq e3 0 1	e7e5
b1c3	rnbqkbnr/pppp1ppp/8/4p3/4P3/8/PPPP1PPP/RNBQKBNR w KQkq e6 0 2	g1f3
b8c6	rnbqkbnr/pppp1ppp/8/4p3/4P3/2N5/PPPP1PPP/R1BQKBNR b KQkq - 1 2	g8f6
f2f4	r1bqkbnr/pppp1ppp/2n5/4p3/4P3/2N5/PPPP1PPP/R1BQKBNR w KQkq - 2 3	f1c4
e5f4	r1bqkbnr/pppp1ppp/2n5/4p3/4PP2/2N5/PPPP2PP/R1BQKBNR b KQkq f3 0 3	e5f4
d2d4	r1bqkbnr/pppp1ppp/2n5/8/4Pp2/2N5/PPPP2PP/R1BQKBNR w KQkq - 0 4	g1f3
d7d5	r1bqkbnr/pppp1ppp/2n5/8/3PPp2/2N5/PPP3PP/R1BQKBNR b KQkq d3 0 4	d8h4
e4d5	r1bqkbnr/ppp2ppp/2n5/3p4/3PPp2/2N5/PPP3PP/R1BQKBNR w KQkq d6 0 5	c1f4
d8h4	r1bqkbnr/ppp2ppp/2n5/3P4/3P1p2/2N5/PPP3PP/R1BQKBNR b KQkq - 0 5	d8h4

- Game Analysis Feature:** Added a robust PGN analysis module allowing users (logged-in or guests) to upload their games and receive move-by-move feedback using Stockfish.
- Chess Tutorials Section:** A dedicated tutorials page was introduced to help users learn chess basics and advanced strategies.

3. **Improved Homepage Interface:** The homepage was redesigned for a cleaner, more professional look aligned with the overall ChessMaster theme.
4. **Enhanced Sidebar Navigation:** Sidebar design and functionality were unified across the application, improving user experience and responsiveness.
5. **Profile Picture Upload:** Users can now upload and display profile photos that reflect across the dashboard and sidebar.
6. **Error Handling & Bug Fixes:** Refined session handling, resolved logout issues, and stabilized multiplayer gameplay logic.
7. **Comprehensive Testing:** Test cases were written for each major functionality and compiled into a structured professional-grade test report.

Future Scope

1. **Interactive Game Analysis with Board Visualization**
Enhance the current PGN analysis by integrating an interactive chessboard. This board will display each move step-by-step, highlight mistakes or blunders, and provide best move suggestions using the Stockfish engine.
2. **Chess Puzzles and Tactics Trainer**
Introduce a feature offering users daily chess puzzles—like mate-in-one, forks, pins, and other tactical exercises—to sharpen their skills in a game-like environment.
3. **Game History Tracking:** Allow users to view, download, and analyze their past games, including win/loss records and move-by-move breakdowns.
4. **Real-Time Multiplayer Chat:** Add chat functionality during multiplayer games to increase interactivity and social engagement.
5. **Mobile Application Version**
Develop a mobile version of ChessMaster for Android (and potentially iOS), enabling users to play, analyze games, and track progress from their smartphones with responsive design and offline features.

GitHub: <https://github.com/Abhishek7Jangir/Chess2.com>

References

1. [Chess.js GitHub Repository](#) – JavaScript chess library used for move validation and PGN handling.
2. Express.js Documentation – Backend web framework used for building the server
<https://expressjs.com/>
3. [Stockfish Chess Engine](#) – Open-source chess engine used for game analysis and AI move generation.
4. [Node.js Official Website](#) – Server-side JavaScript runtime used to build backend services.
5. [Express.js Documentation](#) – Web framework used for routing and middleware integration.
6. [EJS Templating](#) – Templating engine used for rendering dynamic HTML pages.
7. Socket.IO Documentation – Used for real-time multiplayer game communication.
8. [MDN Web Docs](#) – Reference for HTML, CSS, and JavaScript language features.
9. [MySQL Documentation](#) – Used for storing user data, match records, and other backend logic.
10. Socket.IO Documentation – Real-time bidirectional communication library used in multiplayer
<https://socket.io/>